UNIVERSITY COLLEGE OF ENGINEERING

End Semester Examination

COMPUTER ORGANIZATION AND ARCHITECTURE (BCS302)

Roll No. _____

Subject: COMPUTER ORGANIZATION AND ARCHITECTURE

Code: BCS302

Time: 3 Hours

Max. Marks: 70

General Instructions:

1. Attempt all questions.
2. Be precise in your answers.
3. Draw neat diagrams wherever necessary.

SECTION A — SECTION A

Attempt all questions

10 questions × 2 marks each | Attempt all questions

Q.No.	Question	Marks	CO
1	Define the functional units of a digital computer system.	2	CO1
2	What is meant by bus arbitration?	2	CO1
3	State the IEEE 754 standard representation for single-precision floating-point numbers.	2	CO2
4	Define the term 'Arithmetic and Logic Unit' (ALU).	2	CO2
5	What is a micro-operation?	2	CO3
6	Define the purpose of cache memory in a computer system.	2	CO4
7	Define Direct Memory Access (DMA).	2	CO5
8	List the different modes of data transfer between CPU and I/O devices.	2	CO5
9	What is an I/O interface?	2	CO5
10	Define the concept of an interrupt.	2	CO5

SECTION B — SECTION B

Attempt all questions

5 questions × 6 marks each | Attempt all questions

Q.No.	Question	Marks	CO
11	Explain the concept of instruction cycles and its sub-cycles (fetch and execute).	6	CO3

Q.No.	Question	Marks	CO
12	Describe the basic concept of pipelining in processor organization.	**6**	**CO3**
13	Compare 2D and 2 1/2D memory organization of semiconductor RAM.	**6**	**CO4**
14	Discuss the different address mapping techniques used in cache memory.	**6**	**CO4**
15	Explain the working of programmed I/O and interrupt-initiated I/O modes of data transfer.	**6**	**CO5**

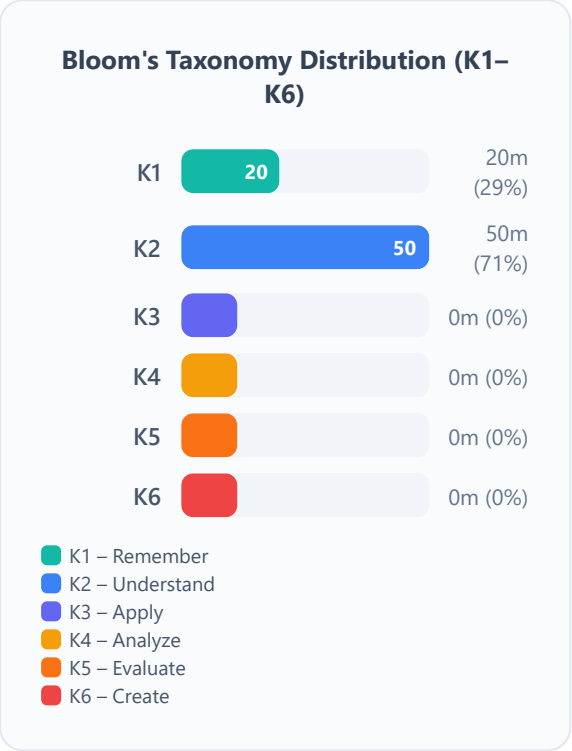
SECTION C — SECTION C

Attempt all questions

2 questions × 10 marks each | Attempt all questions

Q.No.	Question	Marks	CO
16	Describe various addressing modes used in processor organization with suitable examples.	**10**	**CO1**
17	Explain Booth's algorithm for signed operand multiplication with the help of a flowchart.	**10**	**CO2**

Question Paper Analysis



Bloom's Taxonomy Summary		
Level	**Marks**	**%**
K1	20	29%
K2	50	71%
K3	0	0%
K4	0	0%
K5	0	0%
K6	0	0%

Course Outcome Summary		
CO	**Marks**	**%**
CO1	14	20%
CO2	14	20%
CO3	14	20%
CO4	14	20%
CO5	14	20%